

Theory of the Affine-DDPNM

Best Approximation, Nested Decomposition, and Error Separation

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This document develops the Galerkin projection theory of the Affine-DDPNM. Starting from the broken-pressure continuous-traction (C-DD) reference system, we prove: (i) the Schur solution at each enrichment level is the $\hat{\mathbf{S}}^{uu}$ -seminorm best approximation to the full C-DD trace solution; (ii) for nested trace subspaces the error admits an orthogonal Pythagorean decomposition; and (iii) the total volume error satisfies a two-sided bound that separates the C-DD discretisation component from the interface-modal-truncation component, with explicit constants for the A -norm, L^2 -norm, and broken- H^1 seminorm. **Scope.** The reference system throughout is the broken-pressure C-DD formulation (discontinuous P_1 pressure across pores). Equivalence with the monolithic continuous-pressure Taylor–Hood system is not established here; consequently the discretisation term $\|\mathbf{u} - \mathbf{u}_h\|$ appearing in the separation bounds is a C-DD discretisation error, not a standard FEM error. All results assume planar interfaces with a fixed orthonormal frame.

1 C-DD reference system and trace subspaces

1.1 Notation and standing assumptions

We work on a fixed finite-element mesh of the fluid domain $\Omega \subset \mathbb{R}^3$, partitioned into N non-overlapping pore subdomains $\{\Omega_i\}_{i=1}^N$ with m planar internal interfaces $\{\gamma_k\}_{k=1}^m$. On each pore we use the Taylor–Hood $[P_2]^3$ – P_1 pair with **discontinuous** pressure across interfaces (the broken-pressure C-DD formulation). All objects are finite-dimensional; h refers to the given mesh.

Assumption 1.1 (Local algebraic stability). *For every pore i :*

- (a) $A_i \in \mathbb{R}^{n_{u,i} \times n_{u,i}}$ is symmetric positive definite;
- (b) $B_i \in \mathbb{R}^{n_{p,i} \times n_{u,i}}$ has full row rank;
- (c) the constant function 1 belongs to the local pressure space.

Assumption 1.1(a) is guaranteed when the pore has a positive-measure no-slip wall segment.

Assumption 1.2 (Matching interface trace spaces). *For every internal interface γ_k shared by pores i and j , the two local velocity trace spaces coincide with a shared scalar nodal basis $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ and a one-to-one DOF correspondence between the two sides.*

Dimension conventions. m_k = number of scalar trace DOFs on interface k ; $m_\Gamma = 3 \sum_{k=1}^m m_k$ = total C-DD interface DOFs (three components per trace node). For pore i , $r_i^u = 3 \sum_{\gamma \in \mathcal{U}_i} m_{i,\gamma}$ (C-DD internal trace DOFs) and $r_i^k = 3 \sum_{\gamma \in \mathcal{K}_i} m_{i,\gamma}$ (C-DD boundary trace DOFs). $n_i = 9m_i^u + m_i^k$ = affine primitive modes on pore i .

1.2 Interface frames and affine scalar shapes

On interface γ_k fix an orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ directed from the first pore of the pair to the second. Let $\mathbf{c}_k$ be the interface centroid and define the half-span-normalised in-plane coordinates

$$s_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1}}{a_k}, \quad t_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2}}{b_k}, \quad (1)$$

where $a_k, b_k > 0$ are the maximal absolute in-plane coordinate values on γ_k . The three affine scalar shape functions are

$$\varphi_0(\mathbf{x}) \equiv 1, \quad \varphi_s(\mathbf{x}) = s_k(\mathbf{x}), \quad \varphi_t(\mathbf{x}) = t_k(\mathbf{x}), \quad (2)$$

and the mode set is $\mathcal{M} := \{0, s, t\} \times \{n, t_1, t_2\}$.

1.3 C-DD coupling, Schur complement, and energy seminorm

For pore i incident on γ_k , define the oriented inward-traction directions $\mathbf{d}_{i,k,\beta}$ ($\beta \in \{n, t_1, t_2\}$) satisfying $\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}$ for the opposite pore j .

The C-DD coupling block for direction β on γ_k is

$$[N_{i,k}^\beta]_{\ell,j} = \int_{\gamma_k} \psi_{k,\ell} \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} dS \in \mathbb{R}^{m_k \times n_{u,i}}. \quad (3)$$

Collect the three-direction blocks on all internal ports of pore i into $N_i^u \in \mathbb{R}^{r_i^u \times n_{u,i}}$, and on all boundary ports into $N_i^k \in \mathbb{R}^{r_i^k \times n_{u,i}}$. Let $\boldsymbol{\xi}^k \in \mathbb{R}^{m_\Gamma^k}$ (with $m_\Gamma^k = \sum_i r_i^k$) denote the globally prescribed boundary traction nodal values. Through the boundary affine restriction (see the companion mathematical-properties document, Appendix A) these are determined by the prescribed inlet/outlet pressures.

The velocity response operator is $H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}$, with $M_i = B_i A_i^{-1} B_i^\top$. H_i is symmetric positive semidefinite and satisfies $H_i^\top A_i H_i = H_i$, $\text{Ker}(H_i) = \text{Range}(B_i^\top)$.

The **internal-unknown block** of the local C-DD Schur complement is

$$\widehat{\mathbf{S}}_i^{uu} = N_i^u H_i (N_i^u)^\top \in \mathbb{R}^{r_i^u \times r_i^u}. \quad (4)$$

Let $\widehat{R}_i^u : \mathbb{R}^{m_\Gamma} \rightarrow \mathbb{R}^{r_i^u}$ be the Boolean restriction extracting the C-DD DOFs of pore i from the global vector of unknown interface DOFs. The assembled C-DD Schur complement (uu-block) is

$$\widehat{\mathbf{S}}^{uu} = \sum_{i=1}^N (\widehat{R}_i^u)^\top \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u \in \mathbb{R}^{m_\Gamma \times m_\Gamma}. \quad (5)$$

$\widehat{\mathbf{S}}^{uu}$ is symmetric positive semidefinite.

The C-DD right-hand side, with prescribed boundary tractions absorbed, is

$$\widehat{\mathbf{F}} = - \sum_{i=1}^N (\widehat{R}_i^u)^\top N_i^u H_i (N_i^k)^\top \widehat{R}_i^k \boldsymbol{\xi}^k \in \mathbb{R}^{m_\Gamma}, \quad (6)$$

where $\widehat{R}_i^k$ extracts the boundary DOFs of pore i from $\boldsymbol{\xi}^k$.

The **full C-DD system** for the unknown interface DOFs is

$$\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}, \quad \boldsymbol{\xi} \in \mathcal{V} := \mathbb{R}^{m_\Gamma}. \quad (7)$$

Consistency of (7) follows from well-posedness of the underlying broken-pressure Stokes problem; we take it as given. Define the energy seminorm on $\mathcal{V}$ by $\|\boldsymbol{\xi}\|_{\widehat{\mathbf{S}}^{uu}}^2 := \boldsymbol{\xi}^\top \widehat{\mathbf{S}}^{uu} \boldsymbol{\xi}$.

1.4 Restriction operators and nested trace subspaces

Definition 1.3 (Per-interface restriction matrices). For each interface γ_k , define the L^2 -projection vectors $p_k^\alpha = M_k^{-1} b_k^\alpha \in \mathbb{R}^{m_k}$ for $\alpha \in \{0, s, t\}$, where M_k is the face mass matrix on γ_k and $[b_k^\alpha]_\ell = \int_{\gamma_k} \varphi_\alpha \psi_{k,\ell} dS$. For level $\ell \in \{0, 1, 2\}$ with d_ℓ coefficient DOFs per interface,

- $\ell = 0$ (classic P0, $d_0 = 1$): $P_{k,0} = \begin{bmatrix} p_k^0 \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{3m_k \times 1}$;
- $\ell = 1$ (constant-vector P0, $d_1 = 3$): $P_{k,1} = \begin{bmatrix} p_k^0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & p_k^0 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & p_k^0 \end{bmatrix} \in \mathbb{R}^{3m_k \times 3}$;
- $\ell = 2$ (full affine, $d_2 = 9$): $P_{k,2} = \begin{bmatrix} p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 9}$.

Each column occupies only the block-row matching its direction $\beta \in \{n, t_1, t_2\}$.

The global restriction at level ℓ is the block-diagonal assembly

$$\mathcal{P}_\ell := \bigoplus_{k=1}^m P_{k,\ell} : \mathbb{R}^{d_\ell m} \longrightarrow \mathcal{V}. \quad (8)$$

The corresponding **trace subspace** is $\mathcal{W}_\ell := \text{Range}(\mathcal{P}_\ell) \subset \mathcal{V}$.

Lemma 1.4 (Nested trace subspaces). $\mathcal{W}_0 \subset \mathcal{W}_1 \subset \mathcal{W}_2 \subset \mathcal{V}$.

Proof. Each $P_{k,0}$ (resp. $P_{k,1}$) is obtained from $P_{k,1}$ (resp. $P_{k,2}$) by selecting a subset of its columns. The block-diagonal structure of $\mathcal{P}_\ell$ preserves the inclusion of ranges. $\square$

Remark 1.5 (Coefficient-space nesting is not literal). *The coefficient spaces $\mathbb{R}^{d_0 m}$, $\mathbb{R}^{d_1 m}$, $\mathbb{R}^{d_2 m}$ are Euclidean spaces of different dimensions and cannot be written as subsets of one another. The genuine nested structure lives at the level of trace subspaces $\mathcal{W}_\ell \subset \mathcal{V}$. All results below are stated in terms of these trace subspaces. When we refer to “level ℓ ” we mean the subspace $\mathcal{W}_\ell$ and its associated restriction operator $\mathcal{P}_\ell$.*

2 Best approximation in the energy seminorm

For a fixed level ℓ , the **reduced Schur complement** and **reduced right-hand side** are

$$\mathbf{S}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{S}}^{uu} \mathcal{P}_\ell \in \mathbb{R}^{d_\ell m \times d_\ell m}, \quad \mathbf{F}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{F}}. \quad (9)$$

Lemma 2.1 (Reduced consistency). *If the full C-DD system (7) is consistent, then for every level ℓ , $\mathbf{F}_\ell \in \text{Range}(\mathbf{S}_\ell)$.*

Proof. Let $\boldsymbol{\xi}$ be any solution of $\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}$. Set $A_\ell = (\widehat{\mathbf{S}}^{uu})^{1/2} \mathcal{P}_\ell$; then $\mathbf{S}_\ell = A_\ell^\top A_\ell$ and $\mathbf{F}_\ell = A_\ell^\top (\widehat{\mathbf{S}}^{uu})^{1/2} \boldsymbol{\xi}$. Finite-dimensional linear algebra gives $\text{Range}(A_\ell^\top A_\ell) = \text{Range}(A_\ell^\top)$, hence $\mathbf{F}_\ell \in \text{Range}(\mathbf{S}_\ell)$. $\square$

Consequently there exists at least one solution $\boldsymbol{\lambda}_\ell \in \mathbb{R}^{d_\ell m}$ of the reduced system $\mathbf{S}_\ell \boldsymbol{\lambda}_\ell = \mathbf{F}_\ell$. Define the corresponding C-DD trace approximation

$$\boldsymbol{\xi}_\ell := \mathcal{P}_\ell \boldsymbol{\lambda}_\ell \in \mathcal{W}_\ell \subset \mathcal{V}. \quad (10)$$

Let $\boldsymbol{\xi} \in \mathcal{V}$ be any fixed solution of the full C-DD system $\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}$.

Theorem 2.2 (Galerkin best approximation). *Under Assumptions 1.1 and 1.2, for each level $\ell \in \{0, 1, 2\}$,*

$$\|\xi - \xi_\ell\|_{\widehat{\mathbf{S}}^{uu}} = \min_{\mu \in \mathbb{R}^{d_\ell m}} \|\xi - \mathcal{P}_\ell \mu\|_{\widehat{\mathbf{S}}^{uu}}. \quad (11)$$

When $\widehat{\mathbf{S}}^{uu}$ is semidefinite the minimiser need not be unique, but the seminorm value is independent of the choice of representatives for both ξ and λ_ℓ .

Proof. From $\widehat{\mathbf{S}}^{uu} \xi = \widehat{\mathbf{F}}$, $\mathbf{S}_\ell \lambda_\ell = \mathbf{F}_\ell$, $\mathbf{S}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{S}}^{uu} \mathcal{P}_\ell$, and $\mathbf{F}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{F}}$, we obtain

$$\mathcal{P}_\ell^\top \widehat{\mathbf{S}}^{uu} (\xi_\ell - \xi) = \mathbf{0}. \quad (12)$$

Hence $\mathbf{e}_\ell := \xi - \xi_\ell$ is $\widehat{\mathbf{S}}^{uu}$ -orthogonal to $\mathcal{W}_\ell = \text{Range}(\mathcal{P}_\ell)$.

For arbitrary $\mu \in \mathbb{R}^{d_\ell m}$, decompose $\xi - \mathcal{P}_\ell \mu = \mathbf{e}_\ell + \mathbf{v}$ with $\mathbf{v} := \xi_\ell - \mathcal{P}_\ell \mu \in \mathcal{W}_\ell$. By (12), $\mathbf{e}_\ell^\top \widehat{\mathbf{S}}^{uu} \mathbf{v} = 0$, and the Pythagorean identity in the $\widehat{\mathbf{S}}^{uu}$ seminorm yields $\|\xi - \mathcal{P}_\ell \mu\|_{\widehat{\mathbf{S}}^{uu}}^2 = \|\mathbf{e}_\ell\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\mathbf{v}\|_{\widehat{\mathbf{S}}^{uu}}^2 \geq \|\mathbf{e}_\ell\|_{\widehat{\mathbf{S}}^{uu}}^2$. Equality at $\mu = \lambda_\ell$ proves (11).

Semidefinite independence: if $\lambda_\ell, \lambda'_\ell$ both solve $\mathbf{S}_\ell \lambda = \mathbf{F}_\ell$, then $\mathcal{P}_\ell(\lambda_\ell - \lambda'_\ell) \in \text{Ker}(\widehat{\mathbf{S}}^{uu})$, so the seminorm on both sides of (11) is unchanged. Similarly, any two full solutions differ by an element of $\text{Ker}(\widehat{\mathbf{S}}^{uu})$, which leaves the seminorm unchanged (since $\xi_\ell \in \mathcal{W}_\ell$ and $\mathcal{P}_\ell^\top \widehat{\mathbf{S}}^{uu}(\xi_\ell - \xi) = \mathbf{0}$). $\square$

3 Nested orthogonal decomposition

Theorem 3.1 (Orthogonal decomposition for nested trace subspaces). *Under the same assumptions as Theorem 2.2, let $\ell < \ell'$, so that $\mathcal{W}_\ell \subset \mathcal{W}_{\ell'}$ by Lemma 1.4. Then*

$$\|\xi - \xi_\ell\|_{\widehat{\mathbf{S}}^{uu}}^2 = \|\xi - \xi_{\ell'}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\xi_{\ell'} - \xi_\ell\|_{\widehat{\mathbf{S}}^{uu}}^2. \quad (13)$$

Proof. By Theorem 2.2 at level ℓ' , $\mathbf{e}_{\ell'} = \xi - \xi_{\ell'}$ is $\widehat{\mathbf{S}}^{uu}$ -orthogonal to $\mathcal{W}_{\ell'}$. The difference $\xi_{\ell'} - \xi_\ell$ satisfies $\xi_{\ell'} \in \mathcal{W}_{\ell'}$ and $\xi_\ell \in \mathcal{W}_\ell \subset \mathcal{W}_{\ell'}$, hence $\xi_{\ell'} - \xi_\ell \in \mathcal{W}_{\ell'}$. Therefore $\mathbf{e}_{\ell'} \perp_{\widehat{\mathbf{S}}^{uu}} (\xi_{\ell'} - \xi_\ell)$. The decomposition $\xi - \xi_\ell = \mathbf{e}_{\ell'} + (\xi_{\ell'} - \xi_\ell)$ is $\widehat{\mathbf{S}}^{uu}$ -orthogonal; the Pythagorean identity gives (13). $\square$

Corollary 3.2 (Three-level gain decomposition). *For $\mathcal{W}_0 \subset \mathcal{W}_1 \subset \mathcal{W}_2$,*

$$\|\xi - \xi_0\|_{\widehat{\mathbf{S}}^{uu}}^2 = \|\xi - \xi_2\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\xi_2 - \xi_1\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\xi_1 - \xi_0\|_{\widehat{\mathbf{S}}^{uu}}^2. \quad (14)$$

Interpretation of the three right-hand-side terms:

- $\|\xi - \xi_2\|_{\widehat{\mathbf{S}}^{uu}}^2$: **affine residual** — error after the full nine-mode affine approximation;
- $\|\xi_2 - \xi_1\|_{\widehat{\mathbf{S}}^{uu}}^2$: **linear gain** — reduction from adding the six linear modes $\{s, t\} \times \{n, t_1, t_2\}$ to the constant-vector space;
- $\|\xi_1 - \xi_0\|_{\widehat{\mathbf{S}}^{uu}}^2$: **tangential gain** — reduction from adding the two tangential constant modes to the single normal constant mode.

Proof. Apply Theorem 3.1 with $(\ell, \ell') = (0, 1)$ and $(\ell, \ell') = (1, 2)$, then substitute. $\square$

Remark 3.3 (Four-space causal decomposition). *The three-level chain $\mathcal{W}_0 \subset \mathcal{W}_1 \subset \mathcal{W}_2$ lumps together the six linear modes as a single gain term and cannot distinguish whether the dominant missing physics is normal-direction linear variation or tangential-direction linear variation. To resolve this one needs four trace subspaces defined by their per-interface mode content:*

$$\begin{aligned} \mathcal{W}_{0n} &= \text{span}\{1 \cdot n\} \quad (1 \text{ mode/interface: classic } P0), \\ \mathcal{W}_{0v} &= \text{span}\{1 \cdot n, 1 \cdot t_1, 1 \cdot t_2\} \quad (3 \text{ modes/interface: constant vector}), \\ \mathcal{W}_{1n} &= \text{span}\{1 \cdot n, s \cdot n, t \cdot n\} \quad (3 \text{ modes/interface: linear normal only}), \\ \mathcal{W}_{1v} &= \text{span}(\{1, s, t\} \times \{n, t_1, t_2\}) \quad (9 \text{ modes/interface: full affine}). \end{aligned}$$

These four spaces form a commutative diamond: $\mathcal{W}_{0n} \subset \mathcal{W}_{0v} \subset \mathcal{W}_{1v}$ and $\mathcal{W}_{0n} \subset \mathcal{W}_{1n} \subset \mathcal{W}_{1v}$, but $\mathcal{W}_{0v} \not\subset \mathcal{W}_{1n}$ and $\mathcal{W}_{1n} \not\subset \mathcal{W}_{0v}$. Pythagorean decompositions along the two distinct nested paths give two orthogonal partitions of the classic-P0 error:

$$\begin{aligned}\|\boldsymbol{\xi} - \boldsymbol{\xi}_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2 &= \|\boldsymbol{\xi} - \boldsymbol{\xi}_{1v}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\boldsymbol{\xi}_{1v} - \boldsymbol{\xi}_{0v}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\boldsymbol{\xi}_{0v} - \boldsymbol{\xi}_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2, \\ &= \|\boldsymbol{\xi} - \boldsymbol{\xi}_{1v}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\boldsymbol{\xi}_{1v} - \boldsymbol{\xi}_{1n}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\boldsymbol{\xi}_{1n} - \boldsymbol{\xi}_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2.\end{aligned}$$

The first path asks: after adding all three constant-vector modes (tangential gain), how much more does full affine add (linear gain)? The second path asks: after adding linear normal modes (normal-direction enrichment), how much more do tangential modes add? Comparing the per-interface contributions from the two paths resolves whether the error is dominated by missing normal-direction variation or missing tangential traction. This extension to four spaces and the associated two-path decomposition is left to future work.

4 Velocity reconstruction and trace error identity

4.1 Velocity reconstruction from C-DD trace data

For a C-DD trace vector $\boldsymbol{\zeta} \in \mathcal{V}$ (unknown internal DOFs) and prescribed boundary traction data $\boldsymbol{\xi}^k$, the local velocity field on pore i is

$$U_i(\boldsymbol{\zeta}; \boldsymbol{\xi}^k) = -H_i \left[(N_i^u)^\top \widehat{R}_i^u \boldsymbol{\zeta} + (N_i^k)^\top \widehat{R}_i^k \boldsymbol{\xi}^k \right] \in \mathbb{R}^{n_{u,i}}. \quad (15)$$

The first term couples the unknown interface DOFs to the pore interior; the second term injects the prescribed boundary tractions. The global reconstructed velocity field $\mathbf{u}_h(\boldsymbol{\zeta})$ is the discontinuous assembly of the local fields $\{U_i(\boldsymbol{\zeta}; \boldsymbol{\xi}^k)\}$ across pores.

Lemma 4.1 (Trace-to-velocity error identity). *For any two trace vectors $\boldsymbol{\zeta}, \boldsymbol{\zeta}' \in \mathcal{V}$ sharing the same prescribed boundary data $\boldsymbol{\xi}^k$,*

$$\sum_{i=1}^N \|U_i(\boldsymbol{\zeta}) - U_i(\boldsymbol{\zeta}')\|_{A_i}^2 = \|\boldsymbol{\zeta} - \boldsymbol{\zeta}'\|_{\widehat{\mathbf{S}}^{uu}}^2, \quad (16)$$

where $\|U\|_{A_i}^2 := U^\top A_i U$.

Proof. The boundary coupling terms $(N_i^k)^\top \widehat{R}_i^k \boldsymbol{\xi}^k$ are identical for both arguments and cancel in the difference:

$$U_i(\boldsymbol{\zeta}) - U_i(\boldsymbol{\zeta}') = -H_i (N_i^u)^\top \widehat{R}_i^u (\boldsymbol{\zeta} - \boldsymbol{\zeta}').$$

Using $H_i^\top A_i H_i = H_i$,

$$\begin{aligned}\|U_i(\boldsymbol{\zeta}) - U_i(\boldsymbol{\zeta}')\|_{A_i}^2 &= (\boldsymbol{\zeta} - \boldsymbol{\zeta}')^\top \widehat{R}_i^{u\top} N_i^u H_i A_i H_i (N_i^u)^\top \widehat{R}_i^u (\boldsymbol{\zeta} - \boldsymbol{\zeta}') \\ &= (\boldsymbol{\zeta} - \boldsymbol{\zeta}')^\top \widehat{R}_i^{u\top} \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u (\boldsymbol{\zeta} - \boldsymbol{\zeta}').\end{aligned}$$

Summing over i and using (5) gives (16). $\square$

4.2 Notation for the true solution and its approximations

Let $\mathbf{u}$ denote the exact PDE solution of the Stokes problem. Let $\mathbf{u}_h := \mathbf{u}_h(\boldsymbol{\xi})$ be the C-DD full-trace reconstruction using a solution $\boldsymbol{\xi}$ of (7). For level ℓ , let $\mathbf{u}_\ell := \mathbf{u}_h(\boldsymbol{\xi}_\ell)$. All three velocities share the same prescribed boundary tractions $\boldsymbol{\xi}^k$.

Define the following broken norms on the product velocity space $\prod_i H^1(\Omega_i)^3$. For a (possibly discontinuous) velocity field $\mathbf{v}$,

$$\|\mathbf{v}\|_A^2 := \sum_{i=1}^N \nu \int_{\Omega_i} \nabla \mathbf{v} : \nabla \mathbf{v} \, dx, \quad (17)$$

$$\|\mathbf{v}\|_{L^2}^2 := \sum_{i=1}^N \int_{\Omega_i} |\mathbf{v}|^2 \, dx, \quad (18)$$

$$|\mathbf{v}|_{H^1}^2 := \sum_{i=1}^N \int_{\Omega_i} \nabla \mathbf{v} : \nabla \mathbf{v} \, dx = \frac{1}{\nu} \|\mathbf{v}\|_A^2. \quad (19)$$

For a discrete function $\mathbf{v}_h$ with local coefficient vectors $\{V_i\}$, the A -norm satisfies $\|\mathbf{v}_h\|_A^2 = \sum_i V_i^\top A_i V_i$.

5 Volume error—trace error separation

5.1 Two-sided bound: the true error floor

Set

$$D_h := \|\mathbf{u} - \mathbf{u}_h\|_A, \quad E_\ell := \|\mathbf{u}_h - \mathbf{u}_\ell\|_A = \|\boldsymbol{\xi} - \boldsymbol{\xi}_\ell\|_{\mathbb{S}_{uu}}, \quad (20)$$

where the second equality is Lemma 4.1 with $\boldsymbol{\zeta} = \boldsymbol{\xi}$, $\boldsymbol{\zeta}' = \boldsymbol{\xi}_\ell$.

Theorem 5.1 (Two-sided volume-trace bound: A -norm). *Under Assumptions 1.1 and 1.2, for each level ℓ ,*

$$|E_\ell - D_h| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_A \leq E_\ell + D_h. \quad (21)$$

In particular, when the C-DD discretisation error is negligible ($D_h \ll E_\ell$), the total error is essentially the trace error: $\|\mathbf{u} - \mathbf{u}_\ell\|_A = E_\ell + O(D_h)$.

Proof. By the reverse and forward triangle inequalities for the A -norm,

$$|\|\mathbf{u}_h - \mathbf{u}_\ell\|_A - \|\mathbf{u} - \mathbf{u}_h\|_A| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_A \leq \|\mathbf{u}_h - \mathbf{u}_\ell\|_A + \|\mathbf{u} - \mathbf{u}_h\|_A.$$

Substituting E_ℓ and D_h from (20) gives the result. $\square$

Remark 5.2 (Why this is an error floor — and what it does not prove). *On a fixed mesh h , the lower bound $E_\ell - D_h$ shows that the total error cannot drop below the trace-modal-truncation error E_ℓ by more than the C-DD discretisation error D_h . When $D_h \ll E_\ell$, the error on this mesh is trace-dominated: $\|\mathbf{u} - \mathbf{u}_\ell\|_A = E_\ell + O(D_h)$, and further mesh refinement on this particular grid would not help—the limiting factor is the interface subspace $\mathcal{W}_\ell$.*

Caution. *This is a single-mesh statement. Both $E_\ell = E_{\ell,h}$ and D_h depend on h . A mesh-independent asymptotic error floor additionally requires $\liminf_{h \rightarrow 0} E_{\ell,h} > 0$ (or convergence to a positive limit $E_\ell^* > 0$), which is not proved here. If $E_{\ell,h}$ were to decay with mesh refinement (e.g. because the trace space becomes richer on finer meshes), the total error could still decrease. The numerical observation that errors plateau at $\sim 6.5\%$ (planar) and $\sim 14\%$ (watershed) under mesh refinement is empirical evidence for a positive asymptotic floor; its proof requires analysing the h -dependence of the C-DD trace space and the restriction operator $\mathcal{P}_\ell$.*

Computability. $E_\ell = \|\boldsymbol{\xi} - \boldsymbol{\xi}_\ell\|_{\mathbb{S}_{uu}}$ is computable from the full and reduced discrete solutions. $D_h = \|\mathbf{u} - \mathbf{u}_h\|_A$ requires the exact PDE solution $\mathbf{u}$ and is not directly computable; it must be estimated via a manufactured solution, an overkill reference computation, or a separate discretisation error estimator.

5.2 Analogous bounds in L^2 and broken H^1

Theorem 5.3 (Two-sided bound: L^2 -norm). *Under the same assumptions, suppose each pore Ω_i admits a Poincaré–Friedrichs constant $C_{P,i} > 0$ such that $\|\mathbf{v}\|_{L^2(\Omega_i)} \leq C_{P,i} |\mathbf{v}|_{H^1(\Omega_i)}$ for all $\mathbf{v}$ vanishing on the wall portion of $\partial\Omega_i$. Set $C_P := \max_i C_{P,i}$ and define*

$$T_\ell^{L^2} := \|\mathbf{u}_h - \mathbf{u}_\ell\|_{L^2}, \quad D_h^{L^2} := \|\mathbf{u} - \mathbf{u}_h\|_{L^2}. \quad (22)$$

Then the following hold.

- (a) **Two-sided bound with $T_\ell^{L^2}$:** $|T_\ell^{L^2} - D_h^{L^2}| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_{L^2} \leq T_\ell^{L^2} + D_h^{L^2}$.
- (b) **Poincaré upper bound:** $T_\ell^{L^2} \leq \frac{C_P}{\sqrt{\nu}} E_\ell$, and therefore $\|\mathbf{u} - \mathbf{u}_\ell\|_{L^2} \leq D_h^{L^2} + \frac{C_P}{\sqrt{\nu}} E_\ell$.

No corresponding lower bound in terms of E_ℓ is available without a reverse stability estimate (e.g. an inverse inequality bounding the A -norm from above by the L^2 -norm on the discrete velocity space).

Proof. (a) Apply the reverse and forward triangle inequalities for the L^2 -norm to $\mathbf{u} - \mathbf{u}_\ell = (\mathbf{u} - \mathbf{u}_h) + (\mathbf{u}_h - \mathbf{u}_\ell)$. (b) From Lemma 4.1 and per-pore Poincaré–Friedrichs, $(T_\ell^{L^2})^2 = \sum_i \|U_i(\boldsymbol{\xi}) - U_i(\boldsymbol{\xi}_\ell)\|_{L^2(\Omega_i)}^2 \leq \frac{C_P^2}{\nu} \sum_i \|U_i(\boldsymbol{\xi}) - U_i(\boldsymbol{\xi}_\ell)\|_{A_i}^2 = \frac{C_P^2}{\nu} E_\ell^2$. Taking square roots and substituting into the upper bound of (a) gives the second statement. $\square$

Theorem 5.4 (Two-sided bound: broken- H^1 seminorm). *Under the same assumptions, define $T_\ell^{H^1} := |\mathbf{u}_h - \mathbf{u}_\ell|_{H^1}$ and $D_h^{H^1} := |\mathbf{u} - \mathbf{u}_h|_{H^1}$. From (19) and Lemma 4.1, $T_\ell^{H^1} = E_\ell/\sqrt{\nu}$ exactly. Then*

$$|T_\ell^{H^1} - D_h^{H^1}| \leq |\mathbf{u} - \mathbf{u}_\ell|_{H^1} \leq T_\ell^{H^1} + D_h^{H^1} = \frac{1}{\sqrt{\nu}} E_\ell + D_h^{H^1}. \quad (23)$$

Proof. The identity $T_\ell^{H^1} = E_\ell/\sqrt{\nu}$ follows from $|\mathbf{v}|_{H^1}^2 = \|\mathbf{v}\|_A^2/\nu$ and Lemma 4.1. The seminorm triangle inequalities give the two-sided bound. Unlike the L^2 case, both the upper and the lower bound are available here because $T_\ell^{H^1}$ is exactly computable from E_ℓ (no additional inequality needed). $\square$

Remark 5.5 (Pessimism of the L^2 constant). *The constant $C_P = \max_i C_{P,i}$ is the worst-case Poincaré constant across all pores and can be very pessimistic for irregular watershed geometries. In practice the effective constant for the specific error field $\mathbf{u}_h - \mathbf{u}_\ell$ is given by the Rayleigh quotient $\|\mathbf{u}_h - \mathbf{u}_\ell\|_{L^2}/\|\mathbf{u}_h - \mathbf{u}_\ell\|_A$ and can be evaluated numerically.*

6 Discussion

6.1 Summary of the theoretical chain

1. **Best approximation** (Theorem 2.2): the level- ℓ DDPNM trace solution is the $\widehat{\mathbf{S}}^{uu}$ -seminorm projection of the full C-DD trace solution onto $\mathcal{W}_\ell$. This is a standard Galerkin optimality result.
2. **Nested decomposition** (Theorem 3.1, Corollary 3.2): error reductions obtained by enriching the trace subspace are $\widehat{\mathbf{S}}^{uu}$ -orthogonal. The total classic-P0 error decomposes into the affine residual plus two orthogonal gain terms.
3. **Error floor** (Theorems 5.1–5.4): the total volume error satisfies a two-sided bound $|E_\ell - D_h| \leq \|\mathbf{u} - \mathbf{u}_\ell\| \leq E_\ell + D_h$ in each of the three norms (A , L^2 , broken- H^1). When $D_h \ll E_\ell$, the trace error E_ℓ is the effective error floor.

6.2 Limitations of the current theory and directions for adaptive enrichment

The orthogonal gain decomposition (Corollary 3.2) provides the algebraic foundation for residual-driven adaptive enrichment. The enrichment algorithm in `ddpnm3d/hierarchy.py` (`run_adaptive_hierarchy`) implements a two-phase loop: starting from Level 0 on all interfaces, residual indicators (from `ddpnm_core/estimate.py`) drive Dörfler marking to upgrade selected interfaces to Level 1, then Level 2.

What the algorithm converges to. The enrichment operates on a fixed mesh h with a fixed set of three trace subspaces $\mathcal{W}_0 \subset \mathcal{W}_1 \subset \mathcal{W}_2$. Because the state space is finite (3^m possible level assignments), the algorithm terminates in at most $2m$ steps; the limit is the all-Level-2 solution ξ_2 , with error floor $E_2 = \|\xi - \xi_2\|_{\mathbb{S}_{uu}}$. The total error *cannot* reach zero for three independent reasons:

1. **Modal truncation:** the nine affine modes per interface span a strict subspace of the full C-DD trace space (~ 100 nodal DOFs per interface in 3D); the complement is never enriched.
2. **C-DD $\neq$ PDE:** the reference system uses broken pressure; even the full C-DD solution u_h does not equal the monolithic Taylor–Hood solution (let alone the PDE solution u).
3. **Geometric error (watershed only):** per-facet normal dispersion on faceted interfaces introduces an additional representation gap that the fixed-frame affine space cannot close.

Overcoming these would require, respectively, higher-order modal enrichment ($P_2, P_3, \dots$), a proof of C-DD–monolithic equivalence (or a monolithic reference theory), and per-facet normal frames for non-planar interfaces.

Toward a contraction theorem. For a Dörfler-driven loop to provably reduce the error by a fixed factor per iteration ($E_{k+1} \leq qE_k$, $q < 1$), the following ingredients are needed beyond what is established here:

1. **Reliability:** each residual indicator η_i must satisfy $\eta_i^2 \geq c_{\text{rel}} g_i^2$, where $g_i^2 = \|\xi_{\text{enriched}} - \xi_{\text{current}}\|_{\mathbb{S}_{uu}}^2$ restricted to interface i (the local contribution to the orthogonal gain term in Corollary 3.2).
2. **Efficiency:** $\eta_i^2 \leq C_{\text{eff}} g_i^2$ (the indicator does not over-estimate the available gain).
3. **Estimator reduction** or a **saturation** property linking the marked set to the global error reduction.

None of these are proved in the present work; the indicators in `estimate.py` are currently validated only empirically.

Summary of what is and is not proved. Theorem 2.2 (best approximation) and Theorem 3.1 (orthogonal decomposition) are rigorous within the C-DD framework. Theorem 5.1–5.4 provide two-sided volume-trace separation bounds that define the error floor on a fixed mesh. What is missing for a full adaptive convergence theory is: reliability/efficiency of the indicators, a contraction or estimator-reduction theorem, the four-space diamond decomposition (Remark 3.3), and the bridge from the C-DD reference system to either the monolithic FEM or the PDE.

6.3 What is not yet proved

1. **C-DD $\neq$ monolithic FEM.** The C-DD reference system uses discontinuous pressure across pores; the monolithic Taylor–Hood system uses globally continuous P_1 pressure. Equivalence of the two velocity solutions is not established. Consequently the discretisation error D_h appearing in the separation bounds is a C-DD discretisation error, not directly comparable to standard FE error estimates.
2. **Mesh-independent error floor.** The two-sided bounds in Theorems 5.1–5.4 hold on a fixed mesh. Proving that $E_{\ell,h}$ remains bounded away from zero as $h \rightarrow 0$ (the asymptotic floor observed numerically at $\sim 6.5\%$ and $\sim 14\%$) requires an analysis of the h -dependence of the trace space and the restriction operator $\mathcal{P}_\ell$ (Remark 5.2).

3. **Reliability and efficiency of residual indicators.** The link between the orthogonal gain terms and the per-interface residual indicators computed in `ddpnm_core/estimate.py` is currently empirical. Upper and lower bounds (efficiency and reliability à la Babuška–Rheinboldt) are required for a contraction theorem.
4. **Contraction theorem for adaptive enrichment.** Even with reliable and efficient indicators, a Dörfler marking strategy additionally requires an estimator-reduction or saturation property to guarantee $E_{k+1} \leq qE_k$ with $q < 1$.
5. **Four-space causal decomposition.** The three-level chain $\mathcal{W}_0 \subset \mathcal{W}_1 \subset \mathcal{W}_2$ does not separate normal-direction linear enrichment from tangential-direction linear enrichment. The four-space diamond (Remark 3.3) with its two-path Pythagorean decomposition is defined but not yet implemented or numerically evaluated.
6. **Non-planar interfaces.** All results assume planar interfaces with a fixed orthonormal frame. Watershed partitions produce faceted interfaces whose per-facet normal variation introduces additional geometric error not captured by the present analysis.
7. **Higher-order modal enrichment.** Overcoming the modal truncation error floor would require enriching beyond the nine affine modes (e.g. P_2 polynomial shapes $\{1, s, t, s^2, st, t^2\} \times \{n, t_1, t_2\} = 18$ modes/interface). The nested decomposition theory extends naturally to any finite chain of nested trace subspaces, but the construction and numerical validation of P_2 restriction operators is left to future work.

References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.
- [2] A. Toselli and O. Widlund. *Domain Decomposition Methods—Algorithms and Theory*. Springer, 2005.
- [3] R. A. Horn and C. R. Johnson. *Matrix Analysis*, 2nd ed. Cambridge University Press, 2012.